Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 3

Semester Test 3

Module EBN122

Elektrisiteit en Elektronika

November 2016

Module EBN122

Electricity and Electronics

November 2016

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Exam information: **Eksamensinligting:**			
Maximum marks: Maksimum punte:	45	*Full marks:* Volpunte:	46
Duration of paper: Duur van vraestel:	*90 minutes* 90 minute	*Open / closed book:* Oopboek / toeboek:	*Closed* Toe
Total number of pages (including this page): Totale aantal bladsye (hierdie blad ingesluit):		13	

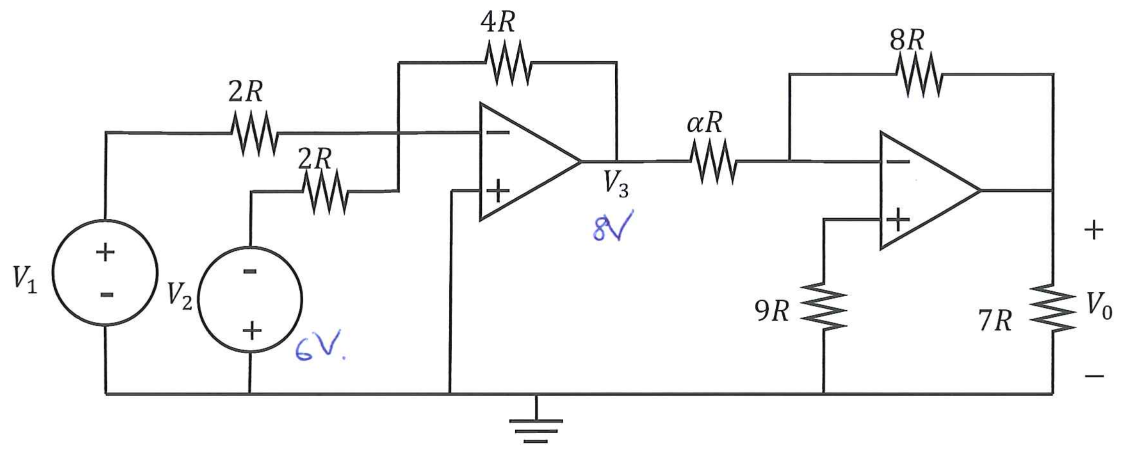
IMPORTANT **- BELANGRIK**

1. *The examination regulations of the University of Pretoria apply.*
   Die eksamenregulasies van die Universiteit van Pretoria geld.
2. *All questions must be answered on the ANSWER SHEET.*
   Alle vrae moet op die ANTWOORDBLAD beantwoord word.
3. *Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
   Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
4. *Answers must include units!*
   Antwoorde moet eenhede insluit!
5. ***Final answers must be written as real numbers, and not as fractions.***
   Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Dosent(e): *Lecturer(s):*	1. D le Roux
	2. R dos Santos
	3. J Joubert

VRAAG/QUESTION 1 [4]

Given: $V_3 = 8$ V and $V_2 = 6$V. a) Calculate V_1. b) Find the value of α for which $V_0 = -16$ V.	Gegee: $V_3 = 8$ V en $V_2 = 6$ V. a) Bereken V_1. b) Vind die waarde vir α sodat $V_0 = -16$ V.



a) $$V_3 = -4R\left(\frac{V_1}{2R} + \frac{(-V_2)}{2R}\right)$$

$$-8 = 2V_1 - 2V_2.$$

$$-8 = 2V_1 - 2(6) \Rightarrow V_1 = +2\text{ V}$$ ✓✓

b) $$V_0 = -\frac{8R}{\alpha R} V_3.$$

$$16 = \frac{8}{\alpha}(8).$$

$$\alpha = \frac{8}{16}(8)$$

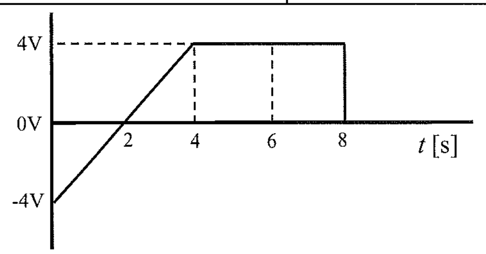
$$\alpha = 4$$ ✓✓.

VRAAG/QUESTION 2 [10]

Question 2.1 [2]	Vraag 2.1 [2]
The voltage across a 2 F capacitor is given as $v(t) = 8 - 3\,e^{-5t}V$. The current through the capacitor at $t\ =\ 0$ is given as 30 A. Determine the current at $t = 2\ s$.	Die spanning oor 'n 2 F kapasitor is $v(t) = 8 - 3\,e^{-5t}V$. Die stroom deur die kapasitor by $t\ =\ 0$ s is 30 A. Bepaal die stroom by $t = 2\ s$.

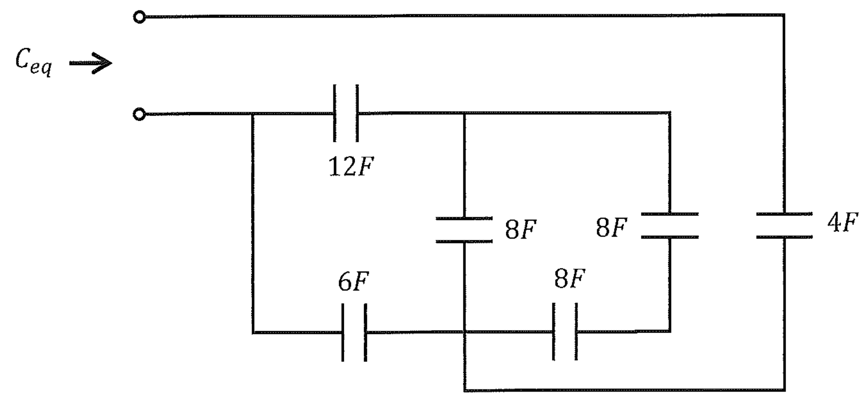
$$i_c = C\frac{dv}{dt}$$
$$= 2\frac{dv}{dt}(8-3e^{-5t})$$
$$= 2(0+15e^{-5t})$$
$$i_c(t) = 30e^{-5t}$$
$$i(2) = 30e^{-10} = 1.36\ mA$$

Question 2.2 [2]	Vraag 2.2 [2]
The graph below shows the voltage across a 10 H inductor. Assuming the initial current through the inductor is 4 A, what is the current between 0 and 6 seconds?	Onderstaande grafiek dui die spanning oor 'n 10 H induktor aan. Indien die aanvanklike stroom deur die inductor 4 A is, wat is die stroom tussen 0 en 6 sekondes?

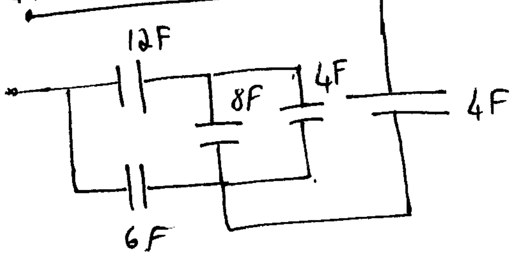


$$i_L(t) = \frac{1}{L}\int_{t_0}^{t_1} v(t) + i(0)$$
$$= \frac{1}{10}\int_0^6 v(t) + 4$$
$$= \frac{1}{10}\int_0^4 v(t) + \frac{1}{10}\int_4^6 v(t) + 4$$
$$= 0 + \frac{1}{10}(4\times 2) + 4$$
$$= 0.8 + 4$$
$$= \underline{4.8A}$$

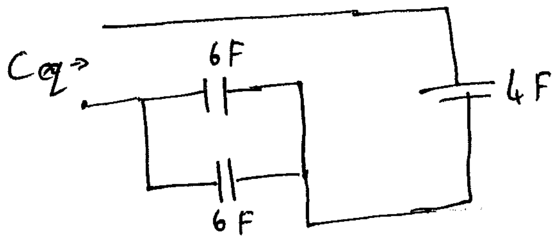
Question 2.3 [2] Calculate C_{eq}.	**Vraag 2.3 [2]** Bepaal C_{eq}.



$8//8 = 4F$

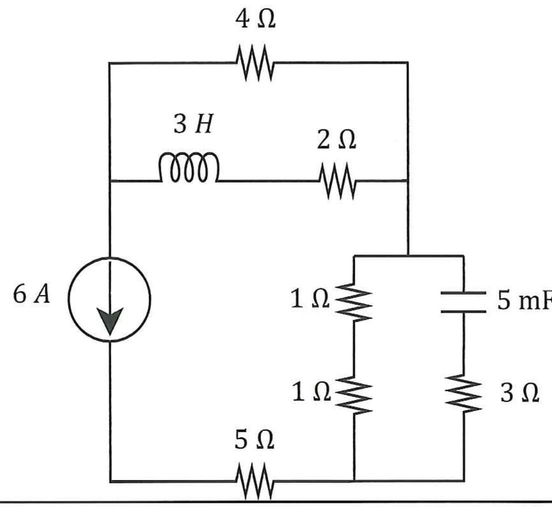


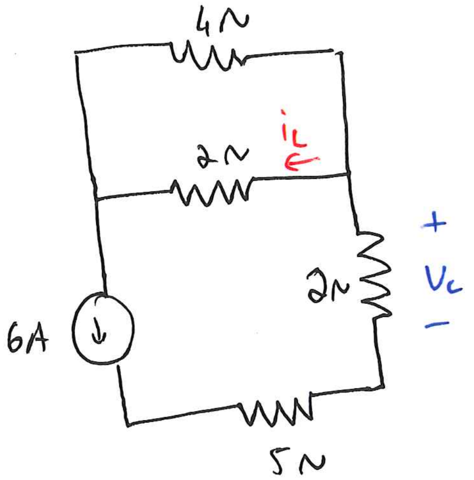
$(8+4)//12 = 6F$



$$C_{eq} = (6+6)//4$$
$$= \frac{12\times 4}{12+4} = 3F$$

Question 2.4 [4] Calculate the energy stored by the capacitor (w_C) and inductor (w_L). Assume direct current conditions.	**Vraag 2.4 [4]** Bepaal die energie gestor in die kapasitor (w_C) en die inductor (w_L). Aanvaar gelykstroom toestande.





$V_c = 6 \times 2 = 12V$

$W_c = \frac{1}{2} C V_c^2$

$= \frac{1}{2}(5 \times 10^{-3})(12)^2$

$= 0.36J$

$= 360mJ$ ✓✓

$i_L = \frac{4 \times 6}{2+4}$ (current division)

$= 4A$

$W_L = \frac{1}{2} L i_L^2$

$= \frac{1}{2}(3)(4)^2$

$= 24J$ ✓✓

VRAAG/QUESTION 3 [16]

Vraag 3.1 [6]	**Question 3.1 [6]**
a) Skryf $v(t)$ as 'n fasor in **reghoekige vorm**: $v(t) = 10cos(\omega t - 90°) - 30sin(\omega t + 45°)$ V b) Bepaal die las-impedansie as $v_L(t) = 300cos(5000\pi t)$ V, en $i_L(t) = 6sin(5000\pi t)$ A. Aanvaar dat die stroom by die + van die verwysingspanning oor die las invloei. Skryf die antwoord in **polêre vorm**. c) 'n 20Ω weerstand en 'n $2\mu F$ kapasitor is in parallel geskakel. 'n Spanningsbron $v(t) = 500cos(5000t + 20°)$ V word oor die parallel-kombinasie aangelê. Bereken die stroom $i_C(t)$ wat deur die kapasitor vloei. Aanvaar dat die stroom vanaf die bron by die + van die verwysingspanning gedefinieer oor die las-impedansie invloei.	a) Write $v(t)$ as a phasor in **rectangular form**: $v(t) = 10cos(\omega t - 90°) - 30sin(\omega t + 45°)$ V b) Determine the load impedance if $v_L(t) = 300cos(5000\pi t)$ V, and $i_L(t) = 6sin(5000\pi t)$ A. Assume the current is flowing into the + of the reference voltage accross the load. Write the answer in **polar form**. c) A 20Ω resistor and a $2\mu F$ kapacitor are connected in parallel. A voltage source $v(t) = 500cos(5000t + 20°)$ V is connected accross the parallel combination. Determine the current $i_C(t)$ that flows through the capacitor. Assume the current is flowing from the source into the + of the reference voltage defined accross the load impedance.

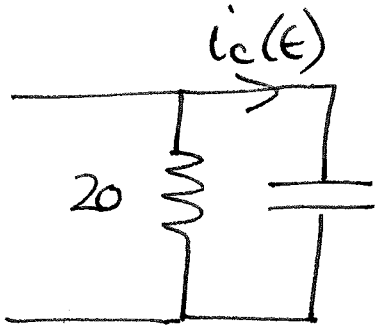
a) $\bar{V} = 10\angle -90° - 30\angle 45° - 90°$

$= -j10 - (21.21 - j21.21)$

$= \underline{-21.21 + j11.21}$

b) $\bar{Z}_L = \frac{300}{6\angle -90°}\ \Omega = \underline{50\angle 90°\ \Omega}$

c)



$\frac{-j}{\omega C} = -j100$

$\bar{I}_c = \frac{500\angle 20°}{100\angle -90°} = 5\angle 110°$ A

$i_c(t) = \underline{5\cos(5000t + 110°)\,A}$

Vraag 3.2 [4]

a) Vereenvoudig en skryf die antwoord in **reghoekige vorm**:

$$\bar{I}_1 = \frac{(3+j4) + 11\angle 90^\circ}{(2\angle 45^\circ)^2 + (3-j4)}$$

b) Bepaal $\overline{V_2}$ in **polêre vorm**:

$$\begin{bmatrix} 2+j4 & -j2 \\ -j2 & 2 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2\angle 90^\circ \end{bmatrix}$$

Question 3.2 [4]

a) Simplify and write the answer in **rectangular form**:

$$\bar{I}_1 = \frac{(3+j4) + 11\angle 90^\circ}{(2\angle 45^\circ)^2 + (3-j4)}$$

b) Determine $\overline{V_2}$ in **polar form**:

$$\begin{bmatrix} 2+j4 & -j2 \\ -j2 & 2 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2\angle 90^\circ \end{bmatrix}$$

a) $$\frac{3+j4+j11}{\cancel{j4}+3-\cancel{j4}} = \frac{3+j15}{3}$$

$$\bar{I}_1 = 1+j5$$

b) $$\Delta = \begin{vmatrix} 2+j4 & -j2 \\ -j2 & 2 \end{vmatrix} = 2(2+j4)-(-j2)(-j2)$$

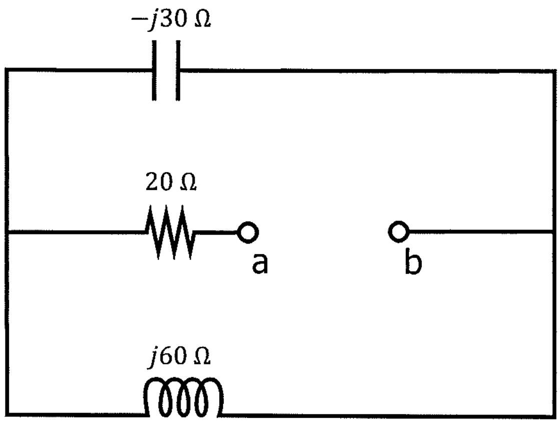
$$= 8+j8$$

$$\Delta_2 = \begin{vmatrix} 2+j4 & 0 \\ -j2 & j2 \end{vmatrix} = j2(2+j4) = -8+j4$$

$$\bar{V}_2 = \frac{-8+j4}{8+j8} = \frac{8.94\angle 153.4^\circ}{11.31\angle 45^\circ}$$

$$= 0.79\angle 108.4^\circ \text{ V}$$

Vraag 3.3 [2] Bepaal die totale impedansie $\bar{Z}_{ab}$ tussen terminale a en b. Skryf die antwoord in **reghoekige vorm**.	**Question 3.3 [2]** Determine the total impedance $\bar{Z}_{ab}$ between terminals a and b. Write the answer in **rectangular form**.

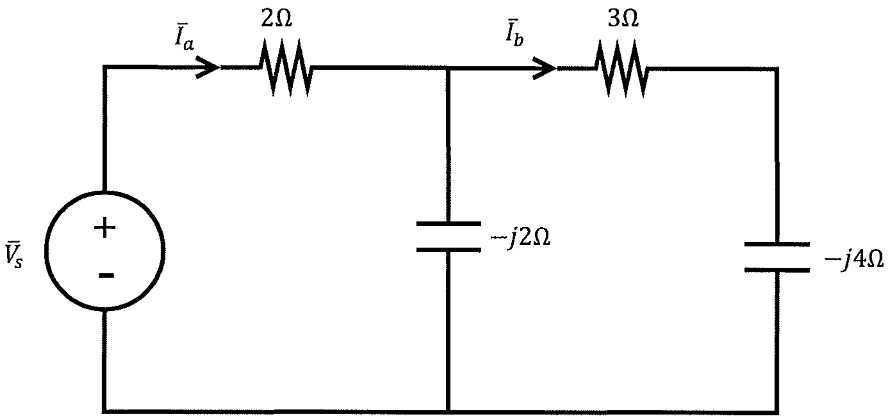


$$\bar{Z}_{ab} = 20 + j60 \,\|\, (-j30)$$

$$= 20 + \frac{(j60)(-j30)}{j30}$$

$$= 20 + \frac{1800}{j30} = \underline{20 - j60\ \Omega}$$

Vraag 3.4 [4]	Question 3.4 [4]
a) Bepaal $\bar{I}_a$ as $\bar{I}_b = 1\angle-90°\ A$. Skryf die antwoord in **reghoekige vorm**. b) Bepaal $\bar{I}_a$ as $\bar{V}_s = 2\angle 60°\ V$. Skryf die antwoord in **polêre vorm**.	a) Determine $\bar{I}_a$ if $\bar{I}_b = 1\angle-90°\ A$. Write the answer in **rectangular form**. b) Determine $\bar{I}_a$ if $\bar{V}_s = 2\angle 60°\ V$. Write the answer in **polar form**.



a) $\bar{I}_a = \bar{I}_b + \dfrac{\bar{I}_b(3-j4)}{-j2}$

$= 1\angle-90° + \dfrac{1\cancel{\angle-90°}\cdot 5\angle-53°}{2\cancel{\angle-90°}}$

$= 1\angle-90° + 2.5\angle-53°$

$= -j + 1.5 - j2 = 1.5 - j3\ A$

b) $Z_T = 2 + (3-j4)\,\|\,(-j2)$

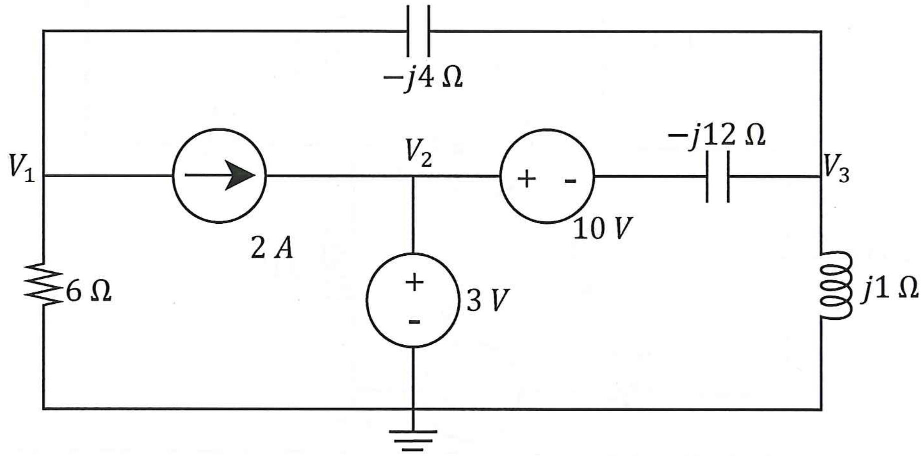
$= 2 + \dfrac{-8-j6}{3-j6}$

$= 2.27 - j1.47 = 2.7\angle-32.9°$

$\bar{I}_a = \dfrac{2\angle 60°}{2.7\angle-32.9°} = 0.74\angle 92.9°\ A$

VRAAG/QUESTION 4 [16]

Question 4.1 [4]	**Vraag 4.1 [4]**
Use nodal analysis to set up two equations of the following form: $(a_1 + ja_2)V_1 + (b_1 + jb_2)V_3 = -24$; $cV_1 + dV_3 = 7$	Gebruik node analise om twee vergelykings van die volgende vorm op te stel: $(a_1 + ja_2)V_1 + (b_1 + jb_2)V_3 = -24$; $cV_1 + dV_3 = 7$



$V_2 = 3V$

Node 1: $2 + \frac{V_1}{6} + \frac{V_1 - V_3}{-j4} = 0.$

$2V_1 + \frac{3V_1}{-j} - \frac{3V_3}{-j} = -24$

$2V_1 + 3jV_1 - 3jV_3 = -24$

$(2+3j)V_1 - 3jV_3 = -24$ ✓ ✓ →

Node 3:

$\frac{V_3 - V_1}{-j4} + \frac{V_3}{j1} + \frac{V_3 + 10 - 3}{-j12} = 0$

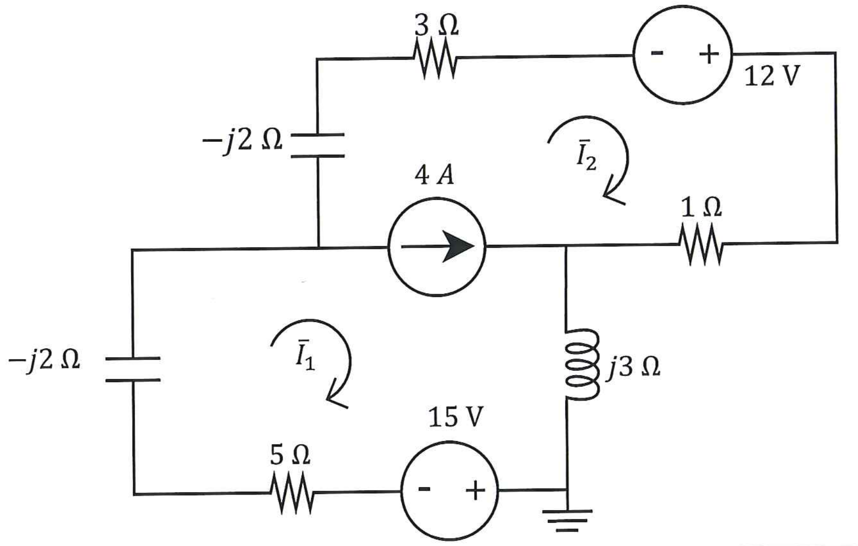
$\frac{jV_3 - jV_1}{4} - jV_3 + \frac{jV_3 + 7j}{12} = 0$

$3jV_3 - 3jV_1 - 12jV_3 + jV_3 + 7j = 0$

$-3V_1 \; \cancel{-} \; 8V_3 = -7$

$3V_1 + 8V_3 = 7$ ✓ ✓ →

Question 4.2 [4] Use mesh analysis to set up two equations of the following form: $(e_1 + je_2)\bar{I}_1 + (f_1 + jf_2)\bar{I}_2 = -3$ $g\bar{I}_1 + h\bar{I}_2 = 4$	**Vraag 4.2 [4]** Gebruik lusanalise om twee vergelykings van die volgende vorm op te stel: $(e_1 + je_2)\bar{I}_2 + (f_1 + jf_2)\bar{I}_3 = -3$ $g\bar{I}_1 + h\bar{I}_2 = 4$

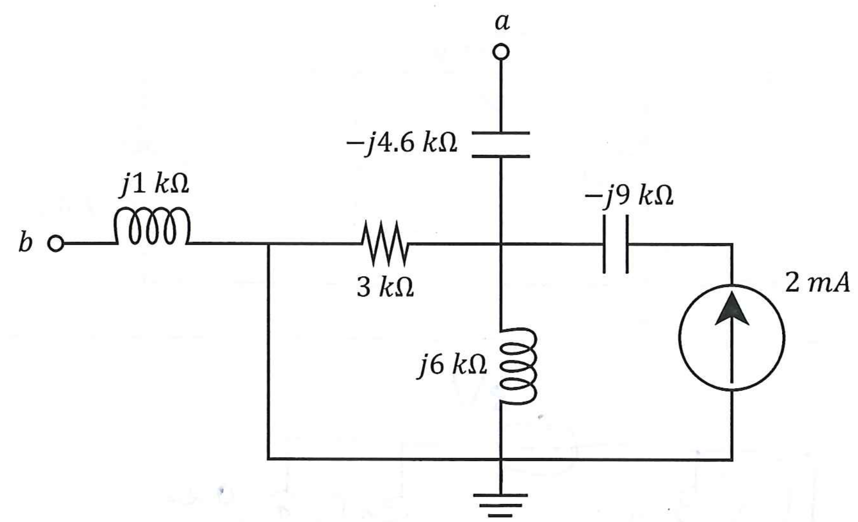


KCL * $\bar{I}_2 + 4 = \bar{I}_1 \Rightarrow 4 = \bar{I}_1 - \bar{I}_2$.

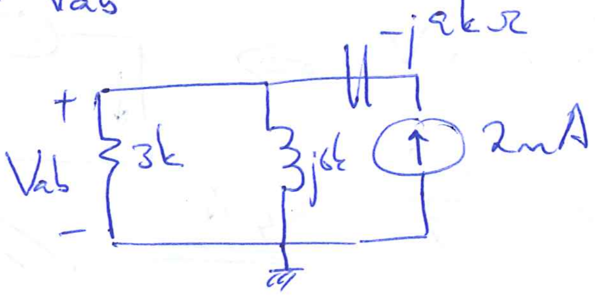
KVL * $(j3 - j2 + 5)\bar{I}_1 + 15 + (-j2 + 3 + 1)\bar{I}_2 - 12 = 0$

$(5+j)\bar{I}_1 + (4-j2)\bar{I}_2 = -3$

Question 4.3 [4] Find the Thevenin equivalent of the circuit with respect to terminals *a*-*b*. a) $\bar{V}_{TH}$ in **rectangular** form b) Z_{TH} in **rectangular** form	**Vraag 4.3 [4]** Vind die Thevenin ekwivalent met verwysing tot terminale. a) $\bar{V}_{TH}$ in **reghoekige** vorm b) Z_{TH} in **reghoekige** vorm.



a) $\underline{V_{TH}} = V_{ab}$

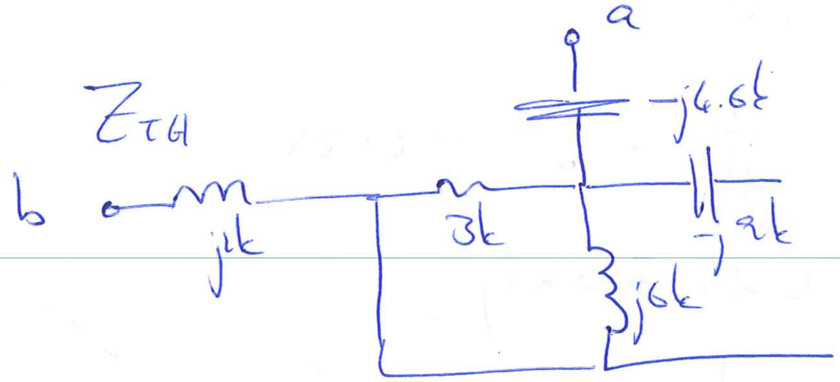


$$\frac{V_{ab}}{3k} + \frac{V_{ab}}{j6k} = 2m$$

$$j2V_{ab} + V_{ab} = j12$$

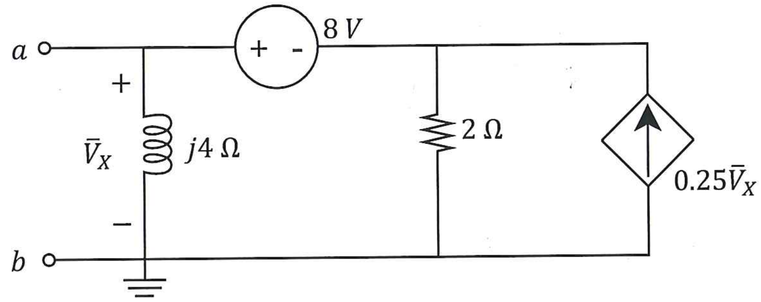
$$V_{ab} = V_{TH} = \frac{j12}{1+j2} = \underline{4.8 + 2.4j\ V}$$

b) Z_{TH}

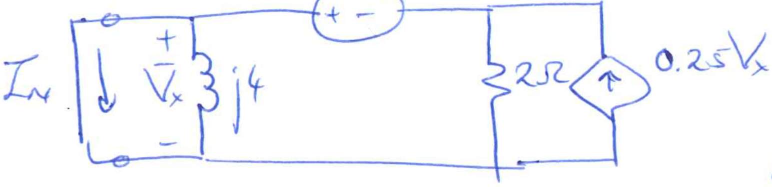


$$Z_{TH} = \left(j1 + (-j4.6) + \frac{3 \times j6}{3+j6}\right)k$$

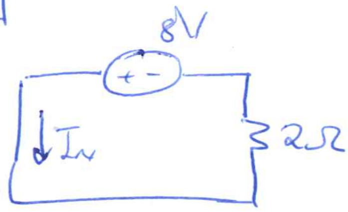
$$= -j3.6k + 2.4k + 1.2jk = (2.4 - j2.4)\ k\Omega$$

Question 4.4 [4] Find the Norton equivalent of the circuit with respect to terminals *a*-*b*. a) $\bar{I}_N$ in **rectangular** form b) Z_N in **polar** form	**Vraag 4.4 [4]** Vind die Norton ekwivalent met verwysing tot terminale *a*-*b*. a) $\bar{I}_N$ in **reghoekige** vorm b) Z_N in **polêre** vorm.
	

$\bar{I}_N$

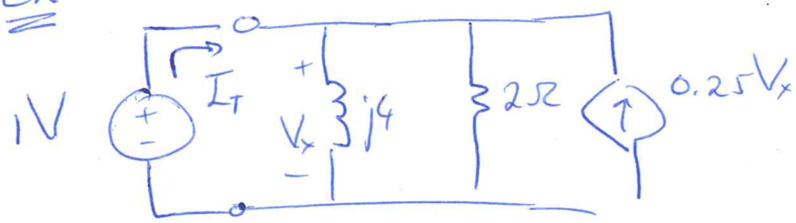


* $\bar{V}_x = 0\,V \Rightarrow$



* $\bar{I}_N = \dfrac{+8}{2} = +4\,A$

Z_N



* $V_x = 1\,V$

* $I_T = \dfrac{1}{j4} + \dfrac{1}{2} - 0.25$

$= 0.25 \mp 0.25j$

* $Z_N = \dfrac{1}{0.25 \mp 0.25j} = 2.828\angle{+45^\circ}\ \Omega.$

EBN122 SEMESTER TEST/ TOETS 2 (NOV 2016)
ANSWER SHEET / ANTWOORDBLAD

Surname: Van:		*Student number:* Studentenommer:									
Initials: Voorletters:		*Group:* Groep:	ENGLISH (Eng 3-1)			ENGLISH (Eng 3-6)			AFR		

Q1+2	14	Q3	16	Q4	16	**Tot:**	46 /45

VRAAG 1 – QUESTION 1 (4)		
1	a) V_1 = 2V	b) α = 4

QUESTION 2 - VRAAG 2 (10)		
2.1	$i_{C(2\,s)}$ = 1.36mA	
2.2	$i_{L(0\ to\ 6\,s)}$ = 4.8A	
2.3	C_{eq} = 3F	
2.4	w_C = 0.36 J = 360mJ	w_L = 24 J

VRAAG 3 – QUESTION 3 (16)		
3.1	a) $\bar{V}$ = −21.21 + j11.21V *(rectangular)*	
	b) $\bar{Z}_L$ = 50∠90° Ω *(polar)*	
	c) $i_c(t)$ = 5 cos(5000t + 110°) A	
3.2	a) $\bar{I}_1$ = 1 + j5 A *(rectangular)*	b) $\bar{V}_2$ = 0.79∠108.4° V *(polar)*
3.3	$\bar{Z}_{ab}$ = 20 − j60 Ω *(rectangular)*	
3.4	a) $\bar{I}_a$ = 1.5 − j3 A *(rectangular)*	b) $\bar{I}_a$ = 0.74∠92.9° A *(polar)*

VRAAG 4 – QUESTION 4 (16)		
4.1	$a_1 + ja_2$ = 2 + j3	$b_1 + jb_2$ = 0 + j(−3)
	c = 3	d = 8
4.2	$e_1 + je_2$ = 5 + j	$f_1 + jf_2$ = 4 + j(−2)
	g = 1	h = −1
4.3	$\bar{V}_{TH}$ = 4.8 + j2.4 V (rectangular)	Z_{TH} = (2.4 − j2.4) kΩ (rectangular)
4.4	$\bar{I}_N$ = 4A (rectangular)	Z_N = 2.828∠45° Ω (polar)